# Deconstructive Analysis of BEN-2293: Molecular Optimization, Genomic Disconnects, and Clinical Failure in AI-Designed Therapeutics

## Executive Summary & Historical Context of AI Candidate Failures

Between 2020 and 2026, artificial intelligence (AI) transitioned from an experimental computational paradigm to a central driver of biopharmaceutical preclinical research, drawing over $100 billion in cumulative global investment[1]. Computational platforms promised to shorten discovery timelines from four years to under eighteen months while systematically de-risking clinical attrition[2]. However, evaluation of clinical pipeline data reveals a persistent translational barrier: as of late 2026, no novel small-molecule or biologic therapeutic designed entirely end-to-end by AI algorithms has achieved full regulatory approval from the U.S. Food and Drug Administration (FDA)[2].

Detailed empirical tracking of AI-derived clinical candidates demonstrates a stark divergence between early- and late-stage performance[2]. Candidates optimized via machine learning achieve Phase I safety and pharmacokinetic success rates of 80% to 90%, substantially outperforming historical industry baselines of 40% to 65%[2]. This high Phase I transition rate demonstrates the proficiency of generative chemistry engines, graph neural networks, and automated ADMET (absorption, distribution, metabolism, excretion, and toxicity) scoring tools at generating stable, highly selective, non-toxic molecules[2].

| Clinical Metric / Milestone | Traditional Drug Discovery Baseline | AI-Enabled Discovery Pipeline (2020–2026) | Primary Failure Mechanism / Bottleneck |
|---|---|---|---|
| **Exploratory Phase Duration** | 42–54 months | 8–18 months[2] | N/A (Demonstrated operational acceleration)[2] |
| **Phase I Trial Success Rate** | 40%–65% | 80%–90%[2] | Safety/PK well-managed by in silico ADMET models[2] |

| | | | |
|---|---|---|---|
| **Phase II Trial Success Rate** | ~40% | ~40%[2] | Efficacy failure due to unvalidated human target biology[2] |
| **Overall IND-to-Approval Odds** | ~7.9%–10% | ~9%–18% (Estimated projection)[2] | Lack of translation from static models to disease physiology[1] |

In Phase II clinical trials—where therapeutic efficacy must be demonstrated in symptomatic patient cohorts—AI-derived candidates drop to the historical industry average success rate of approximately 40%[2]. This pattern clarifies the core limitation of contemporary computational drug design: algorithms excel at solving structural chemistry problems but frequently fail to navigate complex human disease biology[2].

The failure of BenevolentAI's lead candidate, BEN-2293, in a Phase IIa trial for atopic dermatitis in April 2023 serves as a landmark case study of this translational gap[3]. Publicized as a flagship example where AI identified both the biological target hypothesis and guided chemical synthesis, the failure of BEN-2293 to hit its primary efficacy endpoints highlighted the risks of relying on computational target discovery without prospective human tissue validation[3].

# Molecular Architecture and Computational Design Paradigm of BEN-2293

BEN-2293 was developed using the proprietary Benevolent Platform™, an integrated computational framework combining biomedical knowledge graphs, deep learning algorithms, and scientific literature large language models[12]. The platform interrogated multidimensional disease datasets to surface novel druggable nodes capable of modulating complex multifactorial pathologies without incurring off-target toxicity[12].

Through bioinformatic analysis, BenevolentAI identified the neurotrophic tyrosine kinase receptor family—comprising TrkA, TrkB, and TrkC (encoded by the genes *NTRK1*, *NTRK2*, and *NTRK3*)—as a primary driver of both cutaneous inflammation and neurogenic pruritus in atopic dermatitis[13]. Nerve growth factor binding to TrkA on cutaneous sensory nerve fibers triggers signal transduction pathways that mediate persistent itch perception, while concurrent activation of TrkB and TrkC by brain-derived neurotrophic factor and neurotrophin-3/4 amplifies localized neurogenic inflammation[15].

To overcome the limitations of single-isoform Trk inhibitors, the Benevolent Platform™ engineered BEN-2293 as a topical pan-Trk antagonist[12]. The molecule was optimized via generative chemistry algorithms to deliver balanced, sub-nanomolar to low-nanomolar inhibitory potencies ($IC_{50}$) across TrkA, TrkB, and TrkC, preventing functional crosstalk or signal redundancy among neurotrophin receptors[15].

| Parameter / Feature | Specific Benchmark / Value | Bioinformatic & Pharmacological Significance |
|---|---|---|
| **Chemical Name / Identifier** | BEN-2293 (also designated BEN-00002293)[15] | Lead clinical-stage small molecule from BenevolentAI pipeline[12] |
| **Primary Biological Target** | Pan-Trk (TrkA, TrkB, TrkC / *NTRK1*, *NTRK2*, *NTRK3*)[15] | Intended to simultaneously block cutaneous itch signaling and neurogenic inflammation[13] |
| **Biochemical Potency ( $\mathrm{IC}_{50}$ )** | Low nanomolar range across TrkA, TrkB, TrkC[16] | Validates high-affinity binding prediction of generative models[13] |
| **Route of Administration** | Topical Ointment (1% formulation)[19] | Designed to maximize local dermal delivery while avoiding systemic pan-Trk side effects[12] |
| **Preclinical Discovery Platform** | Benevolent Platform™ (Knowledge Graph + Target ID LLM)[12] | Automated extraction of gene-disease associations from biological literature and omics data[13] |
| **Target Indication** | Mild-to-Moderate Atopic Dermatitis[11] | Common chronic inflammatory skin disorder with high unmet need for non-steroidal topicals[21] |

Generative chemistry engines successfully optimized BEN-2293 for high local cutaneous permeability, minimal systemic bioavailability, and low risk of off-target inhibition of related kinase families[12]. Preclinical *in vitro* and *in vivo* models confirmed that BEN-2293 dose-dependently suppressed neurotrophin-induced inflammatory cytokine release, validating the molecule's biochemical profile and setting the stage for clinical testing[16].

# Clinical Evaluation and Phase IIa Endpoint Failure (NCT04737304)

BenevolentAI initiated clinical evaluation of BEN-2293 in a First-in-Human, Phase I/IIa proof-of-concept study registered under ClinicalTrials.gov identifier NCT04737304 (EudraCT:

2020-003143-28-NL)[15]. The study was structured as a two-part, randomized, adaptive, double-blind, vehicle-controlled investigation to assess safety, pharmacokinetics, and clinical efficacy in adult patients diagnosed with mild-to-moderate atopic dermatitis[21].
Part A established dose escalation, safety, and cutaneous tolerability across expanding body surface areas[21]. Part B was a 28-day, parallel-group proof-of-concept trial designed to evaluate the therapeutic effect of topical 1% BEN-2293 twice daily against a vehicle control[19]. The study enrolled 91 adult patients (aged 18 to 65) randomized in a 1.17:1 ratio to receive either BEN-2293 (n=49) or vehicle ointment (n=42)[19].

| Clinical Trial Parameter | Protocol Specification (NCT04737304) | Clinical Outcome / Observed Readout |
|---|---|---|
| **Study Design & Blinding** | Phase I/IIa, Adaptive, Double-Blind, Vehicle-Controlled[21] | Completed Part B 28-day treatment phase successfully with high compliance[21] |
| **Sample Size & Stratification** | Total n=91 (Active: n=49; Vehicle: n=42)[19] | Adults 18–65 with mild-to-moderate AD (baseline itch NRS $\geq 4$)[19] |
| **Primary Safety Endpoints** | Adverse events, local skin tolerance, ECG, lab chemistry[21] | **Met**: Well tolerated, no systemic Trk toxicity, low systemic exposure[6] |
| **Secondary Efficacy Endpoints** | Change from baseline in EASI score and Pruritus NRS score at Day 28[3] | **Failed**: No statistically significant superiority over vehicle control[3] |
| **Exploratory Post-Hoc Findings** | Subgroup analysis based on baseline lesion severity (% BSA)[6] | Modest EASI improvement signal in severe subgroup ($\geq 20\%$ BSA, $p = 0.0296$)[6] |
| **Ultimate Program Outcome** | Corporate pipeline prioritization & strategic review[11] | **Discontinued**: Asset deprioritized, leading to major corporate restructuring[11] |

On April 5, 2023, BenevolentAI announced top-line Phase IIa results[3]. BEN-2293 met its primary safety and tolerability endpoints, exhibiting a clean safety profile with no serious

treatment-emergent adverse events or systemic pan-Trk toxicities[6]. However, the drug failed to demonstrate statistically significant superiority over the vehicle control in either secondary efficacy endpoint: the Eczema Area and Severity Index (EASI) score or the pruritus Numerical Rating Scale (NRS) score[3].

Exploratory post-hoc analyses identified a modest therapeutic signal within a subgroup of patients presenting with more severe baseline involvement (defined as Body Surface Area involvement $\geq 20\%$ , achieving $p = 0.0296$ for EASI improvement)[6]. However, because this post-hoc subgroup signal lacked prospective validation, and because the overall intention-to-treat cohort showed no significant benefit, BEN-2293 was deemed clinically non-viable for mild-to-moderate atopic dermatitis[3]. The failure led to immediate asset deprioritization, followed by a corporate restructuring that included laying off approximately 180 employees and delisting from Euronext Amsterdam[10].

# Multi-Omic Disconnect: Genomic, Transcriptomic, and Environmental Mechanisms of Failure

The clinical failure of BEN-2293 illustrates a fundamental challenge in computational biology: a drug candidate can successfully achieve its designed biochemical activity against a target while still failing to alter overall disease physiology[2]. Post-mortem analysis reveals four biological drivers behind the failure of BEN-2293.

## Genomic Heterogeneity and Epidermal Barrier Integrity

Atopic dermatitis is characterized by structural mutations and transcriptomic downregulation of epidermal barrier proteins[27]. Loss-of-function mutations in the filaggrin gene (*FLG*), along with altered expression of involucrin (*IVL*), loricrin (*LOR*), and keratins (*KRT1*, *KRT10*), weaken the stratum corneum, increasing trans-epidermal water loss and allergen penetration[27].

While Trk receptors are expressed on epidermal keratinocytes and sensory nerves[17], Trk signaling does not regulate the structural assembly of the cornified envelope[27]. By targeting TrkA/B/C exclusively, BEN-2293 left underlying epidermal barrier defects unaddressed, allowing allergen exposure and local inflammatory cascades to persist[13].

## Transcriptomic Feedback and Cytokine Hierarchy

Bioinformatic algorithms selected TrkA (*NTRK1*) based on observed transcriptomic correlations, noting that Type-2 inflammatory cytokines like Interleukin-13 (IL-13) upregulate *NTRK1* expression in human cutaneous tissues[17]. However, knowledge-graph models mistook a downstream transcriptional marker for a master therapeutic driver[11].

In human skin lesions, the immunologic cascade is driven primarily by IL-4, IL-13, IL-31, and TSLP signaling through the Janus Kinase (JAK-STAT) axis[28]. Direct inhibition of Trk receptors fails to block IL-4/IL-13-mediated immune activation or IL-31-induced itch signaling, both of which bypass the Trk axis entirely[28]. Consequently, selective pan-Trk blockade was insufficient to overcome the broader cytokine hierarchy driving the disease[11].

## Environmental Triggers and Microenvironmental Heterogeneity

Transcriptomic profiling of patient skin reveals extreme gene expression heterogeneity driven by environmental exposures[28]. Exposure to environmental contact sensitizers, such as methylisothiazolinone, induces distinct responder sub-phenotypes (e.g., ACD-A high-responders versus ACD-B low-responders)[17]. High-responding inflammatory phenotypes demonstrate co-upregulation of *NTRK1*, *IL9*, and *IL24*, alongside downregulation of anti-inflammatory mediators like *IL18* and *IL37*[17].

AI models trained on static cell lines or non-stratified public databases (such as GTEx or GEO) frequently fail to capture these patient-specific microenvironmental variations[1]. Without prospectively stratifying patient cohorts by multi-omic sub-phenotype, BEN-2293 was administered to a broad mild-to-moderate cohort where neurogenic Trk signaling was not the primary driver of symptoms[10].

## Dermal Pharmacokinetics and the Vehicle Masking Effect

Topical trials in mild-to-moderate atopic dermatitis are susceptible to high vehicle response rates[32]. Plain emollient bases repair trans-epidermal water loss, soothe inflamed skin, and reduce mild itch independently of active drug compounds[21]. In NCT04737304, vehicle-treated patients experienced notable baseline improvements in EASI and NRS scores[3].

Because BEN-2293 targeted a secondary neurogenic pathway rather than a primary cytokine driver, its incremental therapeutic effect over vehicle ointment was too small to achieve statistical significance in a non-stratified cohort[3].

| **Biological Axis** | **Validated Master Pathways (e.g., Dupilumab, JAKi)** | **Pan-Trk Hypothesis Axis (BEN-2293)** | **Mechanism of Translational Disconnect** |
|---|---|---|---|
| **Primary Disease Driver** | IL-4Rα / IL-13 signaling via JAK1/JAK2/TYK2[28] | NGF / BDNF / NT-3 binding to TrkA/B/C receptors[15] | Trk activity is a downstream marker rather than a master driver[11] |
| **Epithelial Barrier Interaction** | Suppresses IL-4/13 to restore filaggrin (*FLG*) & loricrin (*LOR*)[27] | No direct regulatory effect on cornified envelope synthesis[27] | Failed to repair stratum corneum defect, leaving barrier vulnerable[27] |
| **Pruritus Mediation** | Direct IL-31 receptor & JAK-STAT signal inhibition[28] | Direct inhibition of sensory nerve TrkA activation[15] | Redundant non-Trk itch pathways (IL-31, histamine, TSLP) remained active[28] |
| **Transcriptomic** | Broad suppression | Focused | Narrow |

| | | | |
|---|---|---|---|
| **Control** | of Th2 inflammatory gene clusters[28] | suppression of neurogenic inflammatory cascade[16] | anti-neurogenic effect overwhelmed by Th2 cytokine network[11] |
| **Susceptibility to Vehicle Effect** | Robust clinical signal overcomes baseline emollient effect[32] | Modest clinical effect masked by emollient hydration[3] | Failed to show statistically significant superiority over vehicle[3] |

# Structural Failure Modes in AI Drug Discovery

Analyzing the failure of BEN-2293 yields critical insights into computational target discovery, corporate strategy, and the ongoing evolution of the biotechnology sector.

## The Bioinformatic Target Validation Gap

The clinical trajectory of BEN-2293 illustrates the challenge of distinguishing correlational markers from causative therapeutic drivers[3]. Modern AI platforms rely heavily on knowledge graphs, scientific literature mining, and differential expression analysis to construct target hypotheses[12]. These systems efficiently surface nodes that co-occur frequently with disease states, such as the upregulation of *NTRK1* in inflamed skin[13].

However, co-expression does not equal therapeutic leverage[10]. Machine learning models trained on observational bioinformatic datasets struggle to differentiate between primary disease drivers and downstream reactive cellular responses[10]. This creates a structural risk in computational discovery: algorithms successfully design high-affinity, non-toxic inhibitors against biological nodes that are valid in vitro but functionally redundant in symptomatic patients[2].

## The Phase I/II Valuation Paradox and Business Model Risks

The market reaction to the BEN-2293 readout exposed a structural vulnerability in early AI biotech business models[11]. During the initial investment cycle, platform biotechs were valued similarly to software companies, leveraging claims of compressed discovery timelines to command premium valuations[2]. To validate their underlying software loops, companies rapidly advanced proprietary lead assets into clinical development[11].

Because Phase I trials focus on safety, tolerability, and pharmacokinetics—parameters that machine learning models effectively optimize—these assets achieved high early success rates, further boosting corporate valuations[2]. However, this positioning turned clinical pipelines into single-asset proxies for entire software platforms[11]. When Phase II efficacy trials failed to validate the target hypothesis, the market adjusted sharply, forcing pipeline pruning, workforce reductions, and corporate restructurings across the sector[10].

## Industry Realignment Around Closed-Loop Virtual Cell Models

The clinical failure of early AI assets led to a structural shift in how computational tools are integrated into pharmaceutical pipelines[1]. The industry has moved away from viewing

generative chemistry as a standalone solution, recognizing that designing molecules faster does not eliminate target validation risks[2]. This evolution is reflected in sector consolidation, such as the merger between Recursion Pharmaceuticals and Exscientia[10].

Modern discovery architectures focus on building closed-loop translational systems[1]. These platforms integrate spatial transcriptomics, high-throughput functional genomics, patient-derived organoid assays, and AI-driven virtual cell models[1]. Rather than relying on static bioinformatic databases, current workflows use automated biological feedback loops to interrogate target causality in living human tissues prior to initiating lead synthesis[1].

# Sector Comparative Benchmark: Clinical Trajectories of AI-Derived Assets

Comparing BEN-2293 with other clinical-stage AI candidates highlights the spectrum of outcomes across computational drug discovery programs[2].

| Candidate Molecule | Developing Organization | Target & Modality | Indication | Highest Stage / Status | Primary Clinical Outcome & Lessons |
|---|---|---|---|---|---|
| **BEN-2293**[13] | Benevolent AI[13] | Topical Pan-Trk Inhibitor[13] | Atopic Dermatitis[11] | Phase IIa Completed[13] | **Discontinued**: Failed Phase IIa EASI/NRS efficacy endpoints vs vehicle[3]. |
| **Rentosertib** (ISM001-055)[2] | Insilico Medicine[2] | Oral TNIK Inhibitor[2] | Idiopathic Pulmonary Fibrosis[2] | Phase III Initiated (Sept 2026)[2] | **Active**: Showed +98.4 mL FVC improvement in Phase IIa[4]; validates end-to-end AI target and molecule selection[2]. |

| | | | | | |
|---|---|---|---|---|---|
| **DSP-1181**[4] | Exscientia / Sumitomo Dainippon[4] | Oral 5-$HT_{1A}$ Receptor Agonist[38] | Obsessive-Compulsive Disorder[4] | Phase I Completed[4] | **Discontinued**: Abandoned in Phase I due to undisclosed study criteria failures[2]. |
| **EXS-21546**[40] | Exscientia[40] | Oral $A_{2A}R$ Antagonist[40] | Solid Tumors / Immuno-Oncology[40] | Phase I/II Pruned[40] | **Discontinued**: Pipeline pruned during corporate portfolio rationalization[14]. |
| **REC-4881**[5] | Recursion Pharmaceuticals[5] | Small-Molecule Allosteric Inhibitor[5] | Familial Adenomatous Polyposis[5] | Phase II Active[5] | **Active**: AI identified novel indication for existing compound; demonstrated polyp reduction[5]. |

Comparing BEN-2293 with Insilico Medicine's Rentosertib illustrates why target selection determines clinical viability[2]. While BEN-2293 targeted a secondary neurogenic pathway in a disease driven by Th2 cytokines[11], Rentosertib targeted TNIK (Traf2- and Nck-interacting kinase), an essential master node regulating fibrotic tissue remodeling in Idiopathic Pulmonary Fibrosis[2].

In its Phase IIa trial, Rentosertib produced a statistically significant improvement in Forced Vital Capacity (+98.4 mL versus -20.3 mL for placebo at 60 mg once daily), advancing into a 320-patient Phase III trial in late 2026[2]. This contrast demonstrates that computational platforms succeed clinically only when applied to biological targets that exert direct, non-redundant control over disease pathogenesis[2].

# Translational Imperatives and Methodological Outlook

The clinical evaluation of BEN-2293 provides clear guidance for the next generation of computational drug discovery programs[2]. Accelerating early-stage chemistry compresses

timelines, but translational success ultimately depends on rigorous biological validation[2].
To improve clinical transition rates, computational discovery programs should incorporate four core methodological practices:

1. **Prioritize Functional Genomics over Bioinformatic Correlation**: Target identification algorithms must move beyond literature co-occurrence and static differential gene expression analysis[10]. Hypothesized targets should be validated using arrayed CRISPR knockout screens and functional phenotypic assays in patient-derived tissues before initiating generative chemistry[1].
2. **Integrate Prospective Multi-Omic Patient Stratification**: Complex inflammatory conditions encompass diverse transcriptomic endotypes[28]. AI platforms should focus on identifying predictive response biomarkers to stratify patient cohorts prior to Phase II entry, ensuring target engagement is evaluated in patients whose disease is actively driven by the targeted pathway[6].
3. **Deploy Virtual Cell Models to Evaluate Pathway Redundancy**: Generative design tools must evaluate candidate molecules within integrated biological network models rather than isolated protein structures[1]. Simulating pathway crosstalk in silico can highlight potential feedback loops and mechanisms of resistance before candidates enter clinical development[1].
4. **Maintain Distinct Validation Gates Across Platform Workflows**: Platform biotechs should decouple early computational chemistry milestones from clinical target validation[2]. Demonstrating high binding affinity and safety in Phase I validates the chemical design loop but should not be conflated with validating biological efficacy in human disease[2].

Adopting these practices will allow computational biopharmaceutical research to move beyond early translational barriers, pairing high-throughput chemical optimization with validated biological targets to improve clinical success rates[1].

## Sources des citations


1. AI in Drug Discovery: Clinical Failures, Regulatory Reality, and the, https://www.researchgate.net/publication/406938117_AI_in_Drug_Discovery_Clinical_Failures_Regulatory_Reality_and_the_Validation_Crisis_Behind_the_Hype
2. Why No AI-Designed Drug Has Been Approved Yet, https://medicaleventsnow.com/medical-insights/ai-drug-discovery-2026-why-no-ai-designed-drug-has-been-approved-yet
3. AI Drug Discovery Has $8.9 Billion in Hype and Zero FDA Approvals, https://www.clinicaltrialvanguard.com/opinion/ai-drug-discovery-has-8-9-billion-in-hype-and-zero-fda-approvals-when-does-the-bill-come-due/
4. AI-Discovered Drugs in Clinical Trials 2026: Full Pipeline, https://intuitionlabs.ai/articles/ai-discovered-drugs-clinical-trials-2026
5. Has AI discovered a real drug? - New Market Pitch, https://newmarketpitch.com/blogs/news/ai-drug-discovery-real-drug
6. Artificial Intelligence in Drug Discovery (2020–2026) - Noah AI, https://www.noah.bio/insight/421b9c9f-c7c6-4b2f-ad23-463b67624a0f

7. Artificial Intelligence in Drug Discovery and Pharmacology, https://www.ijpsjournal.com/article/artificial-intelligence-in-drug-discovery-and-pharmacology-applications-advances-challenges-and-future-perspectives
8. Artificial Intelligence-Assisted Drug Discovery, https://www.aurigeneservices.com/aidd
9. AI-Designed Drugs in Clinical Trials: What 'In Trials' Means, https://hiwavemakers.com/blog/ai-designed-drugs-human-trials-what-it-means-kids/
10. Has AI Changed the Course of Drug Development? Three Years Later, https://divingintogeneticsandgenomics.com/post/has-ai-changed-drug-development-3-years-later/
11. AI Drug Discovery: Learning More from 'Failure Cases' Than ... - note, https://note.com/pharma_insight/n/n50e8733d8e43?hl=en
12. BenevolentAI Interim Results for the Six Months Ended 30 June 2022, https://www.businesswire.com/news/home/20220926005840/en/BenevolentAI-Interim-Results-for-the-Six-Months-Ended-30-June-2022
13. BenevolentAI Annual Report 2022, https://www.benevolent.com/application/files/9816/7939/1282/BenevolentAI_Annual_Report_2022.pdf
14. Learning from 'Failures and Withdrawals' in AI Drug Discovery - note, https://note.com/pharma_insight/n/n7286d0419e9f?hl=en
15. BEN-2293 - Drug Targets, Indications, Patents - Patsnap Synapse, https://synapse.patsnap.com/drug/5b63571d07b846edb2fd94ed1e69ff03
16. PIPE Presentation - AWS, https://bai-13902-s3.s3.eu-west-2.amazonaws.com/media/3216/6863/1111/PIPE_Presentation.pdf
17. IL-9 and IL-24 biomarkers in the transcriptional signature of contact, https://pmc.ncbi.nlm.nih.gov/articles/PMC12698526/
18. Mechanism-based nonopioid analgesic targets - JCI, https://www.jci.org/articles/view/191346
19. Trial of Lead Atopic Dermatitis Drug Reports Unsatisfactory Results, https://www.dermatologytimes.com/view/trial-of-lead-atopic-dermatitis-drug-reports-unsatisfactory-results
20. BenevolentAI announces top-line Phase IIa results for its topical pan, https://www.benevolent.com/news-and-media/press-releases-and-in-media/benevolentai-announces-top-line-phase-iia-results-its-topical-pan-trk-inhibitor-ben-2293-1-mild-moderate-atopic-dermatitis/
21. A First-in-Human, Double-Blind, Randomised, Vehicle-Controlled, https://onderzoekmetmensen.nl/en/node/51714/pdf
22. A First-in-Human PoC Study With BEN2293 in Patients With Mild to, https://clinicaltrials.gov/study/NCT04737304
23. BenevolentAI lead atopic dermatitis drug misses the mark, https://pharmaphorum.com/news/benevolentai-lead-atopic-dermatitis-drug-misses-mark
24. Inconclusive efficacy of BenevolentAI's AD drug leaves the company,

https://www.clinicaltrialsarena.com/news/benevolentai-atopic-dermatitis-ad/
25. A First-in-Human PoC Study With BEN2293 in Patients With Mild to, https://www.domcushnan.com/nhs/trials/ctgov/NCT04737304
26. BenevolentAI (“BenevolentAI”, “the Company” or “the Group ... - AFM, https://www.afm.nl/downloadregisterfile.aspx?type=openbaarmaking-voorwetenschap&enc=7Rpj0BBaMD5lzfwUlyQ9TIe+XbKYS1JE7+GVfXY2PxNqAT3NVNKp3QCQcI2WUb5gS/PYmhvg5EA+E7FpgiengZj0NLAMUDbP/oPnUtKH27VA0/CZGu7E4hLGdYYwwWpdi+Sjz29I1ZYICY7S5fdPPYFKzAR5THiIxJ8UAXozNDaV7DhxAFwwBD+hmfj/mCFI4IZQZ3z86yU6VYizYj4tL+UQlOTgn/8mgpJhmDNCtsYHBB+T5IBLnhl5pw/py/3DCxHuU80L0Wo1q8EapvyDGGXiBC8udaP5HilvKtNEyTS1eLe+5JfBO+gozCF8RZaK
27. School of Science and Technology, https://apothesis.eap.gr/archive/download/f3036ee5-26d7-45f9-a56c-6097cb45afb7.pdf
28. Updated insights into the molecular pathogenesis of canine atopic, https://pmc.ncbi.nlm.nih.gov/articles/PMC12243452/
29. Insights from a Canine Organoid Model of Atopic Dermatitis - MDPI, https://www.mdpi.com/1422-0067/27/5/2211
30. Neurotrophic tyrosine kinase receptor, type 1 (NTRK1) confers, https://www.researchgate.net/figure/Neurotrophic-tyrosine-kinase-receptor-type-1-NTRK1-confers-epithelial-cell_fig8_268231233
31. IL-9 and IL-24 biomarkers in the transcriptional signature of contact, https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2025.1685396/full
32. Novel Therapeutic Strategies in the Topical Treatment of Atopic, https://pmc.ncbi.nlm.nih.gov/articles/PMC9788207/
33. Comparative Effectiveness and Safety of Novel Topical Agents for, https://www.researchgate.net/publication/394456645_Comparative_Effectiveness_and_Safety_of_Novel_Topical_Agents_for_Atopic_Dermatitis_in_the_Adult_Population_A_Literature_Review
34. sectional and longitudinal transcriptomes ... - Research Square, https://assets-eu.researchsquare.com/files/rs-2006961/v1_covered.pdf?c=1664225787
35. Safety, pharmacokinetics, and efficacy of HY-072808 ointment, a, https://www.researchgate.net/publication/392074238_Safety_pharmacokinetics_and_efficacy_of_HY-072808_ointment_a_novel_PDE4_inhibitor_in_adolescent_and_adult_patients_with_mild-to-moderate_AD
36. Full article: Nerve Growth Factor Signaling and Its Contribution to Pain, https://www.tandfonline.com/doi/full/10.2147/JPR.S247472
37. Decade of AI in Drug Discovery: Hype or Reality?, https://ijpsdronline.com/index.php/journal/article/view/11145
38. Artificial Intelligence in Drug Discovery and Development - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC13288427/
39. Optimizing Pharmaceutical Innovation Through AI: A Pathway to, https://www.pharmacytimes.com/view/optimizing-pharmaceutical-innovation-throu

gh-ai-a-pathway-to-expedited-drug-discovery-and-development
40. EXS-21546 - Drug Targets, Indications, Patents - Patsnap Synapse, https://synapse.patsnap.com/drug/cd13f61769da4605b33420b99d21f0a6
41. The Rise of Machine and Deep Learning in Small-Molecule Drug, https://www.researchgate.net/publication/393754831_Digital_Alchemy_The_Rise_of_Machine_and_Deep_Learning_in_Small-Molecule_Drug_Discovery
42. Artificial Intelligence - LabLife Nordic AB, https://lablifenordic.com/wp-content/uploads/2025/03/CELL-PRESS-SELECTIONS-ARTIFICIAL-INTELLIGENCE.pdf